Building a Simple ETL Pipeline

Components of data pipelines · Extract, Transform, Load · Making it run again tomorrow

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MATH 153 — Data Wrangling / Data Engineering

Today

1. What a pipeline is
2. Extract
3. Transform
4. Load
5. The whole thing, in one function
6. Protecting it

What a pipeline is

The job

- The Bison Café sends us **one sales file per day**.

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- We have three days.

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- We have three days.
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> And we want it to work again tomorrow, *without us*.

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- The important words are every time it runs.
- A notebook you run once by hand is an **analysis**.
- A pipeline runs tomorrow, and next Tuesday, without you.

The five components

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1	**Source**	where the data comes from	3 daily CSVs from the till

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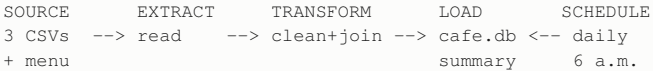
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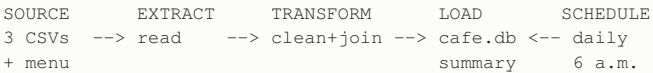
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> Steps 2, 3 and 4 are so common they have a name of their own: ETL.

E — T — L


SOURCE
EXTRACT
TRANSFORM
LOAD
SCHEDULE
3 CSVs
--> read
--> clean+join
--> cafe.db
<-- daily
+ menu
summary
6 a.m.



- **Extract** — get the raw data, *without changing it*.

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- **Transform** — clean it, reshape it, combine it, calculate from it.
- **Load** — write the result somewhere other people can read.

ETL or ELT?

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- **ELT** — load the raw data first, transform it *inside* the database.

> Same three letters, different order. ELT suits very large data and powerful warehouses. ETL suits what we are doing today.

Extract

Look at the data first

```
sale_id,sale_date,sale_time,item,qty,unit_price
S-5001,2027-03-02,14:32,FRUIT CUP,2,$4.00
S-5002,2027-03-02,15:22,Turkey Wrap,2,$7.50
S-5004,2027-03-02,11:58,latte,3,$4.25
```

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```

Before the next click

Two things in this file are already wrong. What are they?

Two problems, both normal

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> Fixing those two things is what “Transform” will mean today.

Extract: read, and change nothing

```python
def extract(data_dir):
    """Read every daily sales file and stack them into one DataFrame."""
    files = sorted(data_dir.glob("sales_2027-*.csv"))
    frames = [pd.read_csv(f) for f in files]
    return pd.concat(frames, ignore_index=True)
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Output

```
107 rows from 3 files
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> **Extract is deliberately boring.** Keeping the raw data untouched means that when something goes wrong later, you know where to start again.

What `glob` is for

```
files = ["sales_2027-03-02.csv", "sales_2027-03-03.csv", ...]  # breaks Friday
files = sorted(data_dir.glob("sales_2027-*.csv"))              # just works
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- The pattern also *excludes* `menu.csv`, which lives in the same folder.

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- `glob` finds files by pattern instead of by name.
- So tomorrow's file is picked up without editing the code.
- The pattern also *excludes* `menu.csv`, which lives in the same folder.
- `sorted` matters: `glob` returns files in filesystem order, not date order.

Transform

Step 1 — make the price a number

```
df["unit_price"] = (df["unit_price"]
                      .str.replace("$", "", regex=False)
                      .astype(float))
```

Output

```
before: $4.00    (a str)
after : 4.0      (a float64)
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> `regex=False` because `$` means “end of string” in a regular expression. Here we want the literal dollar sign.

Step 2 — make the date a real date

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df["sale_date"] = pd.to_datetime(df["sale_date"], format="%Y-%m-%d")
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- You cannot sort it reliably, subtract it, or ask what day of the week it was.
- Convert it *once*, here, and every later step gets a real date.

Step 3 — clean the item names

```
df["item"] = df["item"].str.strip().str.title()
```

Output

```
distinct spellings before: 29
distinct item names after:  8
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- `.str.strip()` removes spaces at the ends — `" Latte "` becomes `"Latte"`.
- `.str.title()` capitalises each word — `"LATTE"` and `"latte"` both become `"Latte"`.

Step 4 — join the menu, then calculate

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df["revenue"] = (df["qty"] * df["unit_price"]).round(2)
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Before the next click

Why did step 3 have to come *before* step 4?

Watch out

`"latte"` would not have matched `"Latte"` in the menu. That sale would have had no category, and its revenue would have quietly vanished from the report.

Two checks after every join

- **Did the row count stay the same?** 107 in, 107 out.

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In practice

Print both numbers. A join that silently doubled your data looks exactly like one that worked.

Load

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> “Load” does not mean one place. It means wherever the people who need the answer will actually look.

Load

```
con = sqlite3.connect("cafe.db")
out.to_sql("sales", con, if_exists="replace", index=False)
con.close()

summary.to_csv("daily_summary.csv", index=False)
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> Dates go into SQLite as `'YYYY-MM-DD'` text, because SQLite has no date type — and in that format, alphabetical order *is* chronological order.

The result

sale_date	category	items_sold	revenue
2027-03-02	Drink	35	$119.75

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Total			**$868.35**

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> 107 sales over three days, from two messy columns and a join.

The whole thing, in one function

run_pipeline()

```python
def run_pipeline(data_dir=DATA, db_path="cafe.db", csv_path="daily_summary.csv"):
    """Extract -> Transform -> Load. Run this every morning."""
    menu  = pd.read_csv(data_dir / "menu.csv")

    raw   = extract(data_dir)                          # E
    clean = transform(raw, menu)                       # T

    assert clean["category"].notna().all(), "some items did not match the menu"
    assert (clean["revenue"] > 0).all(),    "a sale has zero or negative revenue"

    summary = (clean
               .assign(day=clean["sale_date"].dt.date)
               .groupby(["day", "category"], as_index=False)
               .agg(items_sold=("qty", "sum"), revenue=("revenue", "sum"))
               .rename(columns={"day": "sale_date"}))

    load(clean, summary, db_path, csv_path)            # L
    return clean, summary
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    return clean, summary
```

> Twelve lines. This is the deliverable — not the notebook, the function.

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- **It takes arguments.**
- Point it at a different folder and it works on different data.
- **It can be scheduled.**
- A computer can call a function at 6 a.m. It cannot click through a notebook.

Protecting it

Run it twice

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run_pipeline()
run_pipeline()          # on purpose
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Before the next click

How many rows are in the table now?

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rows in the sales table: 107        <-- not 214
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How many rows are in the table now?

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rows in the sales table: 107        <-- not 214
```

> Running it twice gave the same answer as running it once. That property has a name: the pipeline is idempotent.

One keyword makes the difference

```
out.to_sql("sales", con, if_exists="replace", index=False)    # yes
out.to_sql("sales", con, if_exists="append",  index=False)    # NO
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Watch out

With `append`, every run adds another copy. Run it three times and the café has sold everything three times.

This is the single most common bug in a first pipeline.

A log line per step

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TRANSFORM  107 rows cleaned, 8 distinct items
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- Three `print` statements.
- Now when the job runs at 6 a.m. and something looks odd . . .
- . . . you can see which step the rows disappeared at, instead of guessing.

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Where in the pipeline should these two lines go?

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- If a check fails, the pipeline stops *before* writing to the database.
- The manager's report is still yesterday's.
- Which is correct — and much better than today's report being wrong.

Between Transform and Load

- That is the last moment before the numbers reach somebody else.
- If a check fails, the pipeline stops *before* writing to the database.
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In practice

A pipeline that publishes a wrong number is worse than a pipeline that publishes nothing.

Watch a check fail

```
broken = clean.copy()
broken.loc[0, "revenue"] = -5.0          # pretend the transform has a bug

assert (broken["revenue"] > 0).all(), "a sale has zero or negative revenue"
```

Output

```
Pipeline stopped: a sale has zero or negative revenue
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Output

```
Pipeline stopped: a sale has zero or negative revenue
```

> Nothing was written to the database. The bad number never left the building.

The same summary, in SQL

```
SELECT sale_date, category,
       SUM(qty)               AS items_sold,
       ROUND(SUM(revenue), 2) AS revenue
FROM   sales
GROUP  BY sale_date, category
ORDER  BY sale_date, category;
```

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> Once the data is in the database, the report is one query. Same answer, different language.

pandas or SQL?

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> Most real pipelines use both: pandas to clean and load, SQL to query afterwards. That is exactly what we did.

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5. **Put the checks between Transform and Load.**

The one-sentence version

A pipeline is just three functions and a schedule.

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The difference between a pipeline and a notebook
is that somebody else can run it.

In-class activity

Founders Library checkouts

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In practice

Expected: **82 rows**, **16 spellings become 4**, and the table still says 82 after you run it twice.